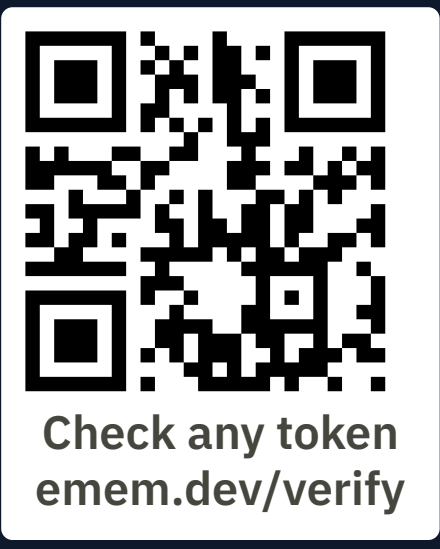


EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Earth-Observation Products, Foundation-Model Embeddings and Signed Execution Traces

Agreement is not evidence. The pixel is.

Every number an agent cites resolves to signed bytes and names the file it came from; open-archive pixels can be re-read.

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THE PROBLEM

Agents that agree are not agents that are right.

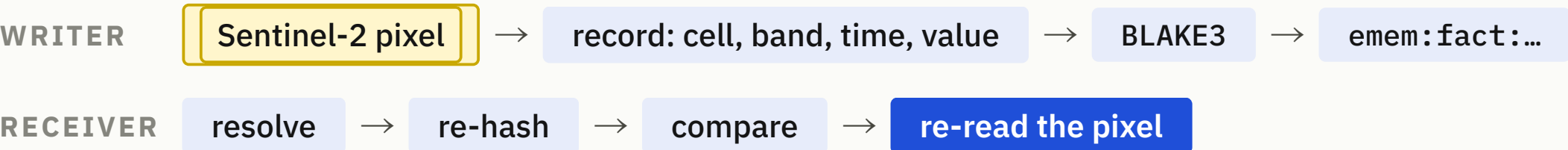
Each summary or handoff re-encodes a reading: the words survive, the referent does not. This is **referential drift**.



Pre-registered compaction study, Gemma-4-12B and Qwen2.5-7B on one host. Under pressure, the 3 pairs that still agree are all wrong.

THE IDEA

Name each satellite observation by the hash of its bytes. **Pass the name, not a sentence about it.**



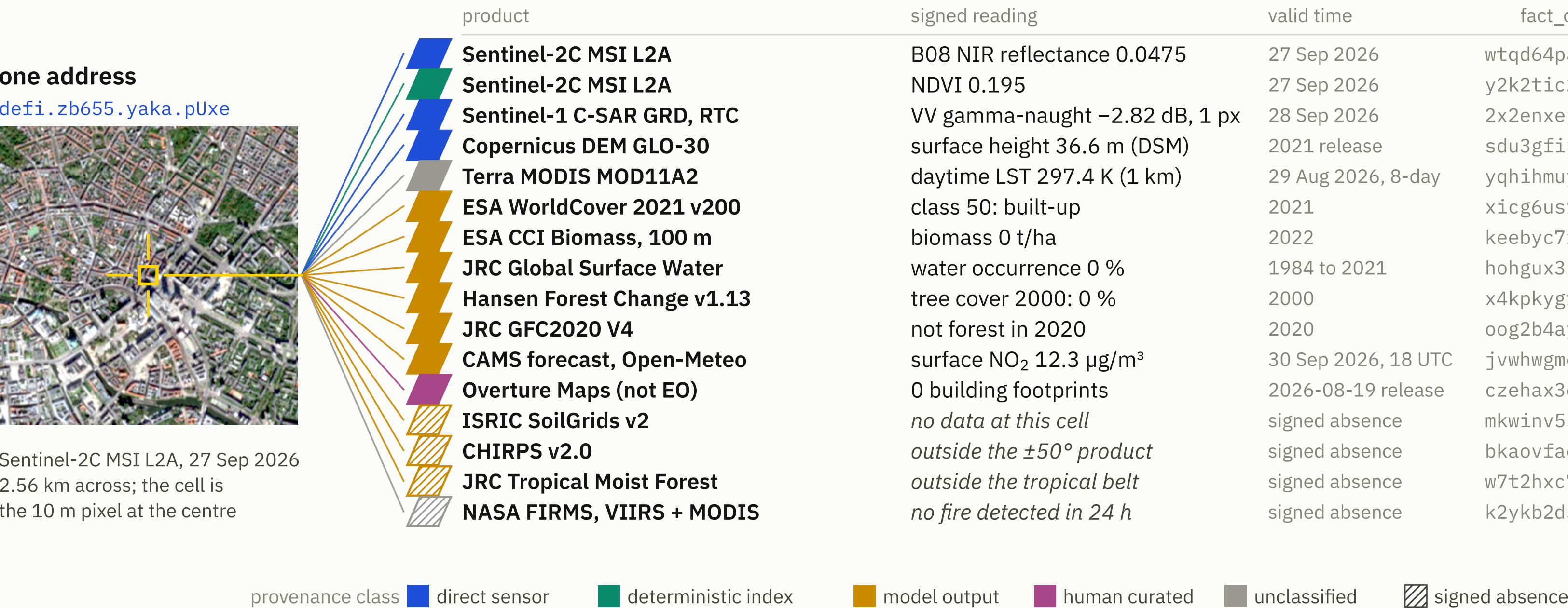
Names move
"the north field" drifts; a 64-bit cell id of about 10 m does not.

Values move
0.4709 becomes "about 0.47"; a fact_cid breaks if one bit changes.

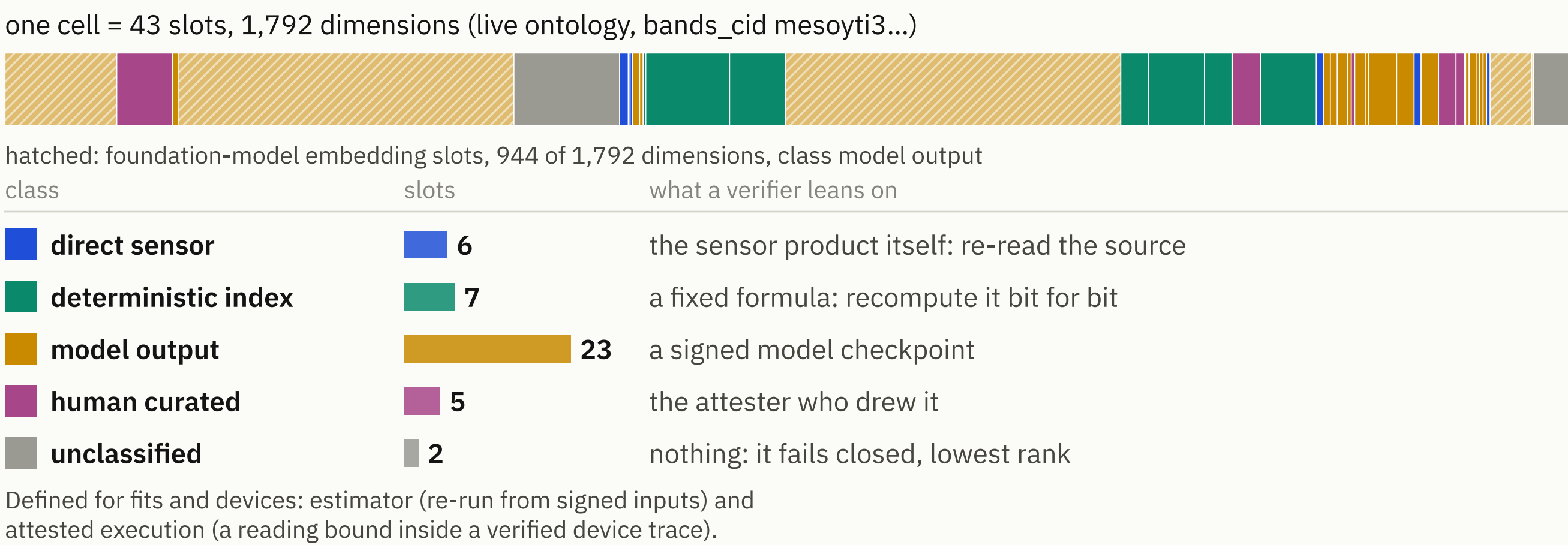
drift $\Delta z = \Delta_{\text{env}} + \Delta_{\text{sensor}} + \Delta_{\text{geo}} + \Delta_{\text{encoder}} + \varepsilon$
world, instrument, misregistration, model, noise: change at one pinned address

1 One address, every product

One 10 m cell in central Berlin, read live on 30 Sep 2026 from each product's native grid (10 m to about 11 km): 15 products, 16 signed facts, 4 of them signed absences.



How emem encodes a memory



Each fact carries a **provenance class**: how far a verifier can check it. **Embeddings are memories too**. Each vector is a model-output fact bound to its encoder checkpoint and recipe, so a new encoder shows up as Δ_{encoder} , not as change on the ground. Every vector emem signed still resolves.

2 The memory model

Six definitions from emem's specification; every panel on this board is an instance of them.

observation $O = (a, b, t, v, u, p, s)$
cell64 address (about 9.55 m), band, valid time, value, uncertainty, provenance, signature

name $\text{cid}(O) = \text{BLAKE3}(\text{CBOR}_{\text{canonical}}(O))$, $s = \text{Ed25519}(\text{BLAKE3}(\text{body}))$
equal values give equal bytes give equal names; one changed bit is a new cid

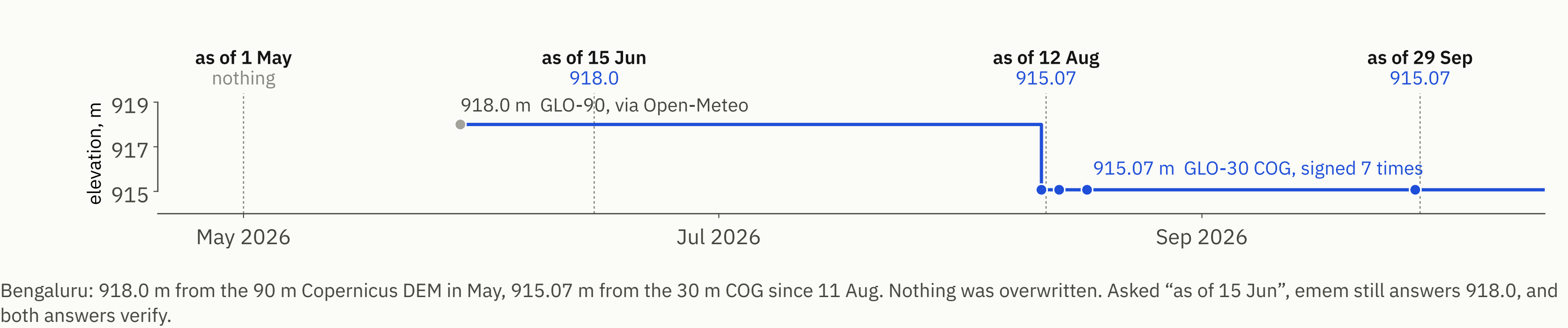
memory $M = (O^*, E^*)$, $E = (\text{subj}, \text{pred}, \text{obj}, t_{\text{from}}, t_{\text{to}})$
append-only; a change is a new observation that supersedes, never an overwrite

recall $\text{recall}(M, a, b \mid t^*, \tau) = \text{latest } O \text{ with } t \leq t^* \text{ and } t_{\text{signed}} \leq \tau$
valid time and transaction time are both inside the signed bytes

recompute $\text{NDVI} = (\text{DN}_8 - \text{DN}_4) / (\text{DN}_8 + \text{DN}_4 + 2 \text{ offset})$
Sentinel-2 L2A digital numbers with the scene's BOA offset (here ~ 1000)

accept $\text{hash} \wedge \text{cell} \wedge \text{Ed25519}(\sigma) \wedge \text{leaf} \in \log \wedge f(\text{DN}) = v \wedge \text{COG}[r, c] = \text{DN}$
checks D to I of section 4, each with stock libraries

Memory on two clocks

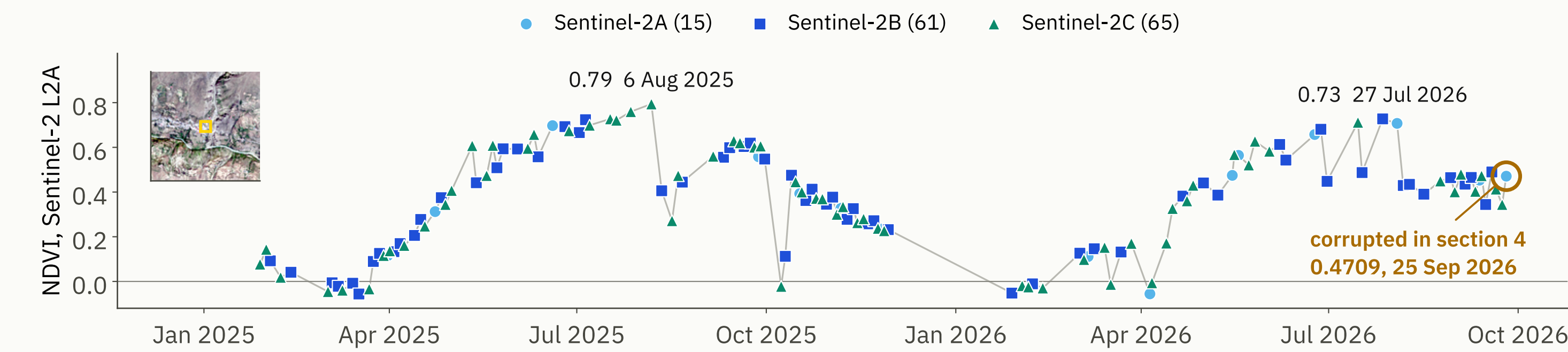


Bengaluru: 918.0 m from the 90 m Copernicus DEM in May, 915.07 m from the 30 m COG since 11 Aug. Nothing was overwritten. Asked "as of 15 Jun", emem still answers 918.0, and both answers verify.

3 Earth observation an agent can cite

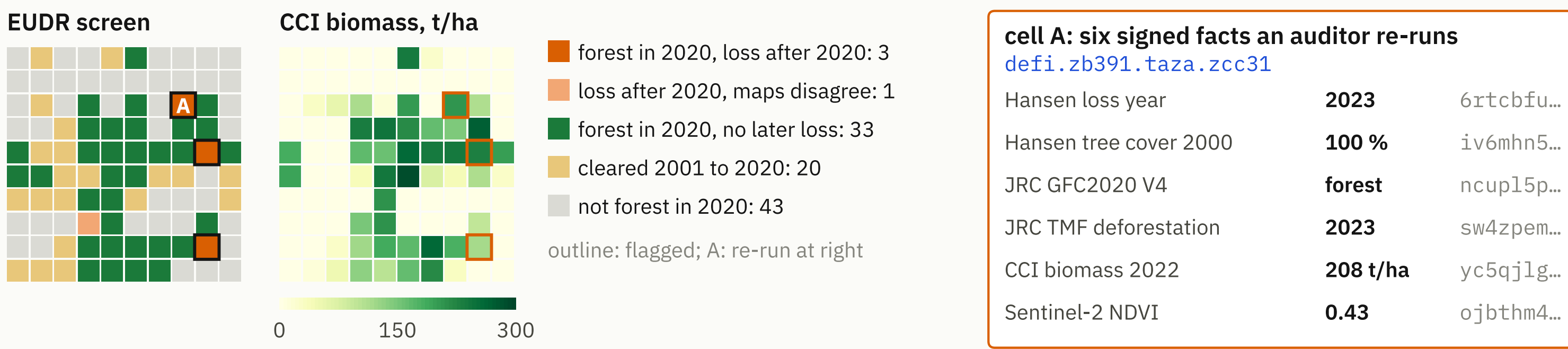
All 780 facts pulled for sections 1 and 3, including the 757 drawn here, were re-hashed, signature-checked and found in the log on 30 Sep 2026.

Two seasons, three satellites, one field



Keylong, Lahaul (India): 141 Sentinel-2 L2A NDVI facts at one 10 m cell, Jan 2025 to Sep 2026. Circled: the record that section 4 corrupts.

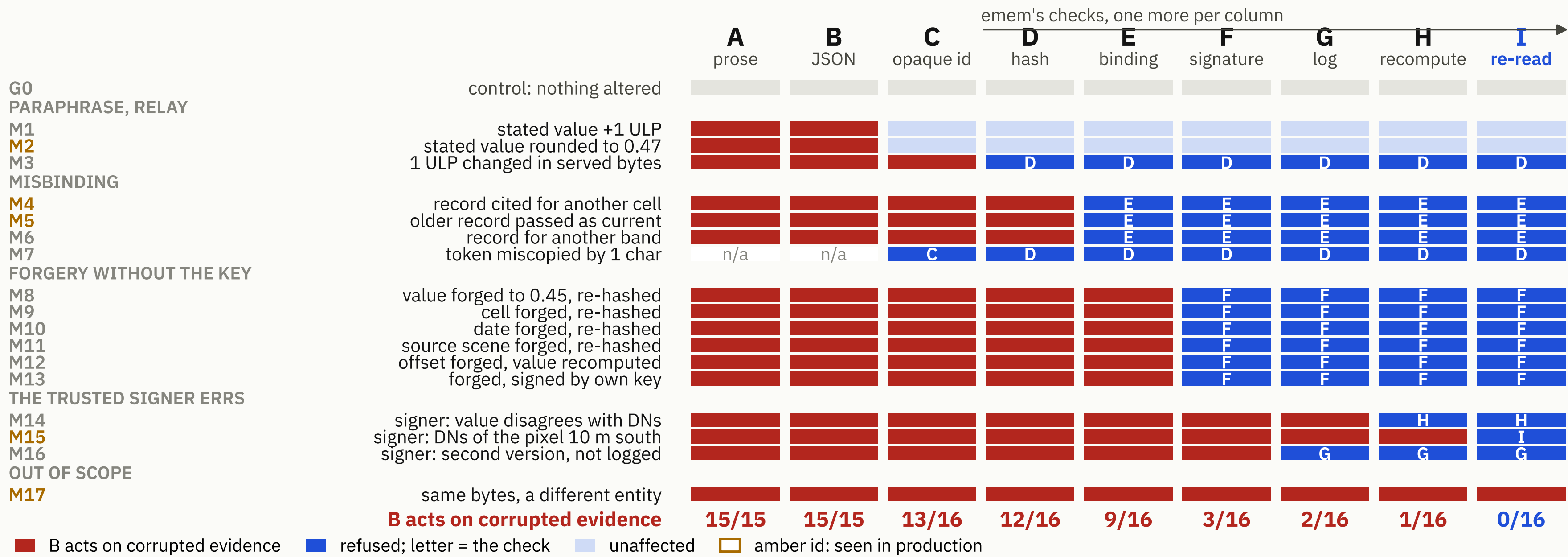
A deforestation screen an auditor can re-run



Rondônia (Brazil): 100 point cells 740 m apart, 600 facts. Flag: forest in 2020, tree-cover loss after 31 Dec 2020. A screen, not a due-diligence statement.

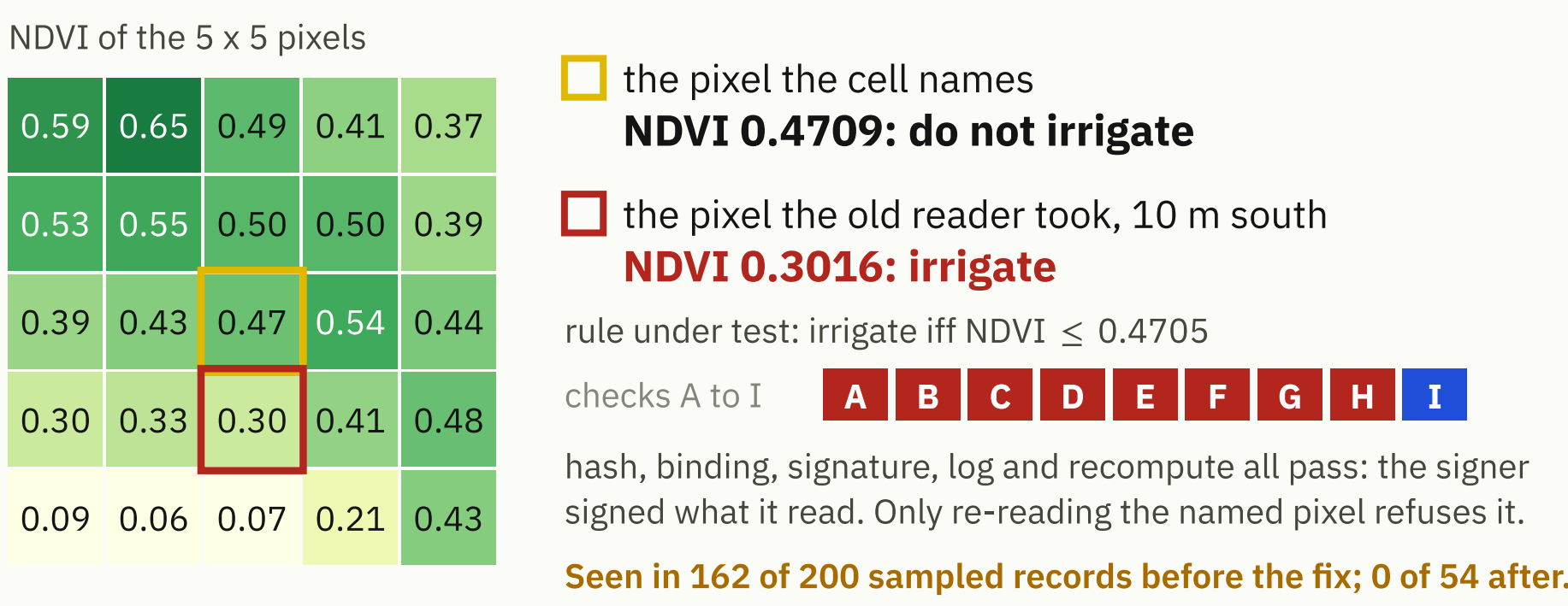
4 Seventeen ways to corrupt one reading

Agent B receives the Keylong record circled in section 3 after one corruption. A deterministic verifier, with no model in the loop, decides whether B acts.



Prose and JSON pass all 15 corruptions they can carry. With all nine checks, none of the 16 in-scope cases gets through; one full check takes 1.2 ms offline, with the source window cached.

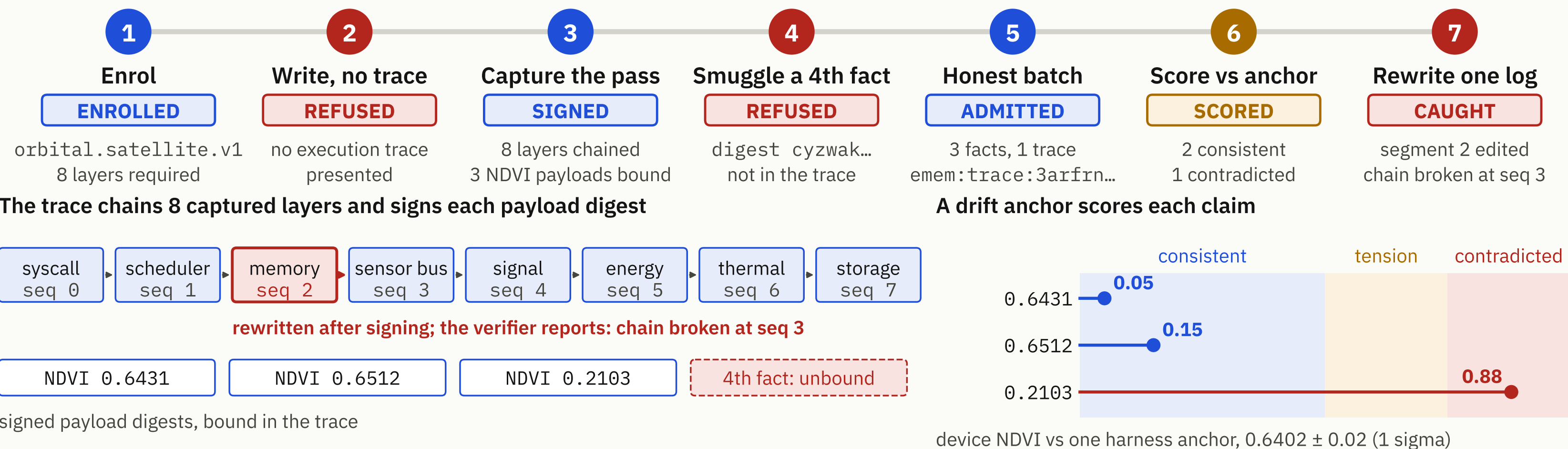
M15 in production: the right record, the wrong pixel



The old reader rounded the pixel centre instead of flooring it, and so read the row below. It signed what it read, so every check up to recompute passes. Before the fix: 162 of 200 sampled records (75 to 86 %, Wilson 95 %). After: 0 of 54.

5 How a satellite could prove what it ran

SAT-042 is a scripted pass in emem's test harness, not a spacecraft; run on 30 Sep 2026.



What a verified token guarantees

INTEGRITY OF	HOW
the source bytes	guaranteed
the observation	bound
the derivation	recomputable
the upstream file	named
the value's accuracy	inherited
the entity	out of scope

Objections, answered

A signature does not make a value true.

Agreed. It fixes which bytes were served. Truth is tested by re-reading the pixel; accuracy comes from the product's own validation.

Why not STAC, openEO or C2PA?

STAC names the scene, openEO the process, C2PA the asset. emem signs the one value an agent cites and names the scene and file it came from.

Is a satellite writing to emem today?

No spacecraft is enrolled. emem.dev re-checks any signed trace from its one enrolled Linux host; SAT-042 exercises every refusal.

1 Agents hand each other a signed observation, not a sentence about it.

2 One address joins every product, so an EO screen can be re-run fact by fact.

3 Only the full chain refuses all 16; M15 needs the pixel re-read.

